

Paper Plan

Endogenous Education Choice in Segmented Search Markets

Ramzi Chariag

March 13, 2026

Core idea. The paper develops and estimates a structural equilibrium model in which workers endogenously choose whether to remain in a low-skill labour market or invest in training that grants access to a higher-skill labour market. Training changes persistent market eligibility and interacts with equilibrium search frictions, endogenous match destruction, job-ladder dynamics in the skilled sector, and aggregate regime changes.

The model is the centrepiece. Empirical exercises demonstrate how the framework interprets labour-market dynamics, training decisions, and policy interventions across two qualitatively different crises.

The paper is organised in three parts.

Part I (Sections 1–4) is conceptual: it presents the model, describes the data, defines the moments, and discusses identification.

Part II (Sections 5–6) is methodological: it explains the multi-regime estimation strategy, defines the parameter classification, and develops the transition-dynamics computation (recovering tightness paths, updating distributions).

Part III (Sections 7–11) is application: the financial crisis, COVID-19, a cross-crisis comparison, followed by two counterfactual policy exercises using the estimated model—Application 1 studies the effects of raising unemployment flow values b_U and b_S on expected unemployment spell durations and match quality, and Application 2 evaluates two education-subsidy policies.

Contents

I	Conceptual Foundations	3
1	Introduction	3
2	Model	4
3	Data	4
4	Moments and identification	5
4.1	Moment list	5
4.2	Identification intuition	5

4.3	Weight matrix	6
II	Estimation and Transition-Dynamics Methodology	6
5	Multi-regime estimation strategy	6
5.1	Parameter classification	6
5.2	Stage 1: baseline estimation	7
5.3	Stage 2: crisis re-estimation	7
5.4	Structural stability check	7
6	Transition dynamics: methodology	7
6.1	The problem	7
6.2	Steady-state comparison vs. transitional path	7
6.3	Recovering tightness along the transition	8
6.4	Updating distributions	8
6.5	The transition map	8
6.6	Implementation	8
III	Applications	8
7	Financial crisis	9
7.1	Baseline estimation (2003–2007)	9
7.2	Crisis estimation (2008–2009)	9
7.3	Training cutoff: financial crisis	9
7.4	Transition dynamics: financial crisis	9
8	COVID-19	10
8.1	Pre-COVID baseline (2015–2019)	10
8.2	Crisis estimation (2020–2021)	10
8.3	Training cutoff: COVID	10
8.4	Transition dynamics: COVID	10
9	Comparing the two crises	10
9.1	Parameter shifts	10
9.2	Training-cutoff comparison	10
9.3	Transition-dynamics comparison	11
10	Application 1: higher b_U and b_S—unemployment spell durations and match quality	11
10.1	Motivation	11
10.2	Exercise design	11
10.3	Outcomes to report	12
10.4	Discussion	12
11	Application 2: education-subsidy policies	13
11.1	Motivation	13
11.2	Key distinction between the two policies	13
11.3	Exercise design	14

11.4 Outcomes to report	14
11.5 Discussion	14
12 Conclusion	15

Part I

Conceptual Foundations

1 Introduction

Purpose.

- Motivate the interaction between business cycles and human-capital decisions (Delas and Sakellaris, 2003; Blom et al., 2021).
- Emphasise that workers respond to labour-market conditions not only through job search but also through retraining and market switching (Clark, 2011; Barr and Turner, 2015).
- Highlight that existing search models rarely incorporate endogenous training decisions with persistent market eligibility (Charlot and Decreuse, 2005; Flinn et al., 2016; Agopsowicz, 2016).
- Argue that the nature of the crisis matters: a demand shock (financial crisis) and a supply/reallocation shock (COVID) affect training incentives through different channels, and a structural model can disentangle them (Acabbi et al., 2024; Leibing, 2026).

Key message. Business cycles influence not only employment outcomes but also the boundary between work and human-capital investment. The channel through which a crisis operates—demand destruction vs. supply disruption—determines whether training incentives rise or fall, with lasting consequences for skill composition (Bičáková et al., 2021, 2023).

Contributions.

1. Develop a structural search model with endogenous training and segmented labour markets, building on the DMP framework (Diamond, 1982; Mortensen and Pissarides, 1994; Pissarides, 2000) and equilibrium wage dispersion models (Postel-Vinay and Robin, 2002; Cahuc et al., 2006).
2. Estimate the model on U.S. labour-market data via SMM, following the structural estimation tradition of Lise et al. (2016) and Lise and Robin (2017).
3. Propose a multi-regime estimation strategy—estimate pre-crisis, fix deep parameters, re-estimate regime-specific parameters during crisis—and apply it to two qualitatively different crises.
4. Study transition dynamics after each type of shock, relating to the cyclical analysis in Shimer (2005) and Moscarini and Postel-Vinay (2024).

5. Evaluate how higher unemployment flow values b_U and b_S affect expected unemployment spell durations and match quality, contributing to the literature on UI generosity and reservation-wage behaviour (Nekoei and Weber, 2017; Schmieder et al., 2016; Lindner and Reizer, 2020).
6. Evaluate education-subsidy policies: compare raising the training flow payoff b_T (stipends) against reducing the cost of education $c(x)$ (investing in higher-education quality), contributing to the literature on education policy in equilibrium (Abbott et al., 2019; Heckman et al., 1998).

2 Model

Purpose. Present the economic environment, derive equilibrium conditions, characterise endogenous cutoffs, and describe transition dynamics after regime changes. (The full model is in the companion model notes.)

Key mechanisms.

- Training grants persistent access to the skilled labour market, as in segmented-market models with endogenous education (Acemoglu, 2001; Charlot and Decreuse, 2005; Charlot et al., 2005).
- Workers decide whether to train when unemployed in the unskilled sector (Charlot and Decreuse, 2010; Charlot and Malherbet, 2013; Decreuse and Granier, 2013).
- Skilled workers face job-ladder dynamics through on-the-job search (Postel-Vinay and Robin, 2002; Cahuc et al., 2006; Bagger et al., 2014; Moscarini and Postel-Vinay, 2016).
- Match destruction occurs endogenously through productivity shocks (Mortensen and Pissarides, 1994; Robin, 2011).
- Aggregate regime changes shift productivity and shock distributions (Lise and Robin, 2017; Moscarini and Postel-Vinay, 2024).

Important model outputs.

- Training cutoff $\bar{x}(z)$
- Separation thresholds $p^*(x, z)$ and $p_S^*(x, z)$
- Market tightness $\theta_U(z)$ and $\theta_S(z)$
- Stationary cross-sectional distributions
- Worker transition dynamics after a regime switch

3 Data

Purpose. Describe the data sources used to construct empirical moments. Full variable-level details are in the companion document *Data and Moments*.

Sources.

- Current Population Survey — Basic Monthly (CPS): unemployment rates, skilled share, training share, transition rates (UE, EU, EE).
- CPS Annual Social and Economic Supplement (ASEC): wage moments.
- Job Openings and Labor Turnover Survey (JOLTS): vacancy counts for market tightness.

Estimation windows. Moments are computed over four distinct periods, each approximating a stationary regime:

Window	Period	Role
Pre-FC baseline	Jan 2003 – Dec 2007	Full estimation
Financial crisis	Jan 2008 – Jun 2009	Crisis re-estimation
Pre-COVID baseline	Jan 2015 – Dec 2019	Stability check + 2nd baseline
COVID crisis	Mar 2020 – Dec 2021	Crisis re-estimation

4 Moments and identification

Purpose. Define the 25 empirical moments, explain what each moment identifies in the model, and describe the SMM weight matrix.

4.1 Moment list

The 25 moments are grouped into four blocks:

Labour-market stocks (9 moments). Overall, unskilled, and skilled unemployment rates; skilled share; training share; wage variance and third central moment by skill group. These pin down the stationary distributions and the training cutoff.

Transition rates (6 moments). Job-finding rates (UE) by skill; EU separation rates by skill; skilled EE rate; training rate. The UE rates identify matching efficiencies and tightness. The EU rates identify shock arrival and match-destruction hazards. The EE rate identifies the job-ladder intensity and helps separate the offer distribution Γ_z from p_S^* (Postel-Vinay and Robin, 2002; Cahuc et al., 2006).

Wages (7 moments). Mean, median, and standard deviation of real hourly wages by skill; log skill premium. Wage levels identify productivity shifters P_U , P_S . The premium identifies the return to training. Dispersion moments identify match-quality distributions and bargaining weights (Lise et al., 2016).

Tightness (3 moments). θ_U , θ_S , and θ_S/θ_U . These directly constrain vacancy costs k_U , k_S through free entry (Pissarides, 2000).

4.2 Identification intuition

- **Deep parameters** are identified from moments that depend primarily on structural features (wage premium $\rightarrow$ type distribution; matching elasticities $\rightarrow$ shape of the JFR–tightness relationship; bargaining weights $\rightarrow$ wage dispersion).
- **Regime-specific parameters** are identified from moments that shift across windows (unemployment rates, tightness, separation rates, JFR levels).

- The training share and training rate pin down the training cutoff $\bar{x}(z)$ and the cost function $c(x)$.
- Higher-order wage moments (variance, skewness) identify the shape of match-quality distributions (Lise et al., 2016; Lise and Robin, 2017).

4.3 Weight matrix

The SMM objective is

$$Q(\theta) = [\hat{m} - m(\theta)]' \hat{\Xi} [\hat{m} - m(\theta)].$$

Initial pass: diagonal $\hat{\Xi} = \text{diag}(1/\hat{\sigma}_1^2, \dots, 1/\hat{\sigma}_K^2)$ using influence-function standard errors.
 Second pass: full optimal $\hat{\Sigma}^{-1}$ if well-conditioned.

Part II

Estimation and Transition-Dynamics Methodology

5 Multi-regime estimation strategy

This section explains the estimation exercises before executing them.

5.1 Parameter classification

Parameters fall into two groups:

Deep structural parameters (constant across all regimes).

- Worker type distribution: a_ℓ, b_ℓ
- Training cost coefficient: a_c
- Matching elasticities: η_U, η_S
- Bargaining weights: β_U, β_S
- OJS flow cost: σ_S

These govern the technology and institutions of the labour market. They should be invariant to the business cycle.

Regime-specific parameters (free within each window).

- Sector productivity shifters: $P_U(z), P_S(z)$
- Matching efficiencies: $\mu_U(z), \mu_S(z)$
- Vacancy costs: $k_U(z), k_S(z)$
- Shock arrival rates: $\lambda_U(z), \lambda_S(z)$

- Damage/offer shape: $\alpha_U(z)$, $a_\Gamma(z)$, $b_\Gamma(z)$
- Exogenous skilled separation: $\xi_S(z)$
- UI flow values: $b_U(z)$, $b_T(z)$, $b_S(z)$

These capture the aggregate state of the economy.

5.2 Stage 1: baseline estimation

Estimate all parameters (deep + regime-specific) by matching the model's stationary equilibrium to pre-crisis moments. The baseline window is the pre-FC period (2003–2007).

5.3 Stage 2: crisis re-estimation

Hold the deep parameters fixed at their Stage 1 values. Re-estimate only the regime-specific parameters by matching crisis-period moments. This is done separately for the financial-crisis window (2008–09) and the COVID window (2020–21).

5.4 Structural stability check

Test whether the deep parameters from Stage 1 (estimated on 2003–07) can rationalise the pre-COVID expansion (2015–19) with only regime-specific parameters re-estimated. This is a powerful out-of-sample validation.

6 Transition dynamics: methodology

This section develops the methodology for computing transition paths after an unexpected regime change in general, before applying it to any specific crisis.

6.1 The problem

Consider an unexpected, permanent switch from z_0 to z_1 at calendar time T . At T^+ , regime-dependent primitives jump to their z_1 values, but the cross-sectional distributions are predetermined (inherited from the old steady state). Equilibrium objects such as $\theta_U(t)$ and the training cutoff $\bar{x}(t)$ must be recomputed using the current transitional distributions. The transition is therefore a joint forward equilibrium path.

6.2 Steady-state comparison vs. transitional path

Two distinct objects:

- (i) **Stationary-regime cutoffs.** $\bar{x}^{ss}(z_0)$ vs. $\bar{x}^{ss}(z_1)$: how the long-run training rule differs across regimes.
- (ii) **Transitional cutoff.** $\bar{x}(t)$ evolves jointly with tightness and distributions: $\bar{x}^{ss}(z_0) \rightarrow \bar{x}(T^+) \rightarrow \bar{x}(t) \rightarrow \bar{x}^{ss}(z_1)$. The path may be non-monotone.

6.3 Recovering tightness along the transition

At each date t , market tightness is pinned down by free entry given the current composition of job seekers (Pissarides, 2000):

Unskilled market.

$$q_U(\theta_U(t)) = \frac{k_U(z_1)}{\int_0^1 J_U(x, 1, t) u_U(x, t) dx}.$$

Skilled market.

$$q_S(\theta_S(t)) = \frac{k_S(z_1)}{J_S(t)},$$

where $J_S(t)$ is the expected value of a contact with a random job seeker drawn from the current pool of unemployed and on-the-job searchers.

Inversion. $\theta_j(t) = (\mu_j/q_j(t))^{1/\eta_j}$. Done at each time step.

6.4 Updating distributions

Given current tightness and policy rules, advance the distributions one step forward:

$$\partial_t t(x, t) = \tau(x, t) u_U(x, t) - (\varphi + \nu) t(x, t), \quad (1)$$

$$\partial_t u_U(x, t) = \nu \ell(x) + \delta_U(x, t) e_U(x, t) - [f_U(t) + \tau(x, t) + \nu] u_U(x, t), \quad (2)$$

$$\begin{aligned} \partial_t u_S(x, t) = & \varphi t(x, t) + [\xi_S + \delta_S^{\text{end}}(x, t)] e_S^{\text{tot}}(x, t) \\ & - [\nu + f_S(t)(1 - \Gamma_{z_1}(p_S^*(x, t)))] u_S(x, t). \end{aligned} \quad (3)$$

6.5 The transition map

$$(u_U, t, u_S, e_S)_t \implies (\theta_U, \theta_S)_t \implies (U_U^{\text{search}}, U_U, T, U_S, \dots)_t \implies (\tau, p_U^*, p_S^*, p_S^{oj})_t \implies \partial_t(u_U, t, u_S, e_S).$$

6.6 Implementation

Forward simulation with within-date equilibrium updating:

1. Start from the stationary distribution of regime z_0 .
2. Switch primitives to z_1 .
3. At each date, solve within-date equilibrium given current distributions.
4. Advance distributions by Δt .
5. Repeat until convergence to the stationary equilibrium of z_1 .

Part III

Applications

7 Financial crisis

7.1 Baseline estimation (2003–2007)

Execute Stage 1: estimate all parameters on pre-FC moments. Report parameter estimates, model fit, and moment-by-moment comparison.

7.2 Crisis estimation (2008–2009)

Execute Stage 2a: fix deep parameters, re-estimate regime-specific parameters on financial-crisis moments.

Expected findings.

- P_U, P_S fall (aggregate productivity decline).
- μ_U, μ_S fall (matching efficiency drops amid uncertainty) (Shimer, 2005).
- k_U, k_S may rise (credit tightening raises vacancy-posting costs).
- λ_U rises (more frequent match destruction) (Mortensen and Pissarides, 1994).
- b_U roughly stable (standard UI was not dramatically expanded).

7.3 Training cutoff: financial crisis

Compute $\bar{x}^{ss}(z_{FC})$ and compare to $\bar{x}^{ss}(z_0)$.

Hypothesis. The training cutoff falls (fewer workers train): both skilled and unskilled prospects worsen roughly symmetrically in a demand shock, and the option value of waiting is high (Dellas and Sakellaris, 2003; Barr and Turner, 2015).

7.4 Transition dynamics: financial crisis

Simulate the transition path from the pre-FC steady state to the financial-crisis regime. Plot the time paths of:

- Unemployment rates $U_U(t), U_S(t)$
- Training stock $T(t)$
- Market tightness $\theta_U(t), \theta_S(t)$
- Training cutoff $\bar{x}(t)$
- Skilled share
- Mean wages by sector

8 COVID-19

8.1 Pre-COVID baseline (2015–2019)

Execute structural stability check: fix deep parameters from the FC baseline, re-estimate regime-specific parameters on pre-COVID moments. Report fit.

8.2 Crisis estimation (2020–2021)

Execute Stage 2b: fix deep parameters, re-estimate regime-specific parameters on COVID-crisis moments.

Expected findings.

- P_U falls sharply (contact-intensive unskilled sector hit hardest).
- P_S falls less (remote-work compatibility).
- ξ_S spikes (mass temporary layoffs).
- b_U, b_T rise substantially (FPUC, PUA expanded UI benefits) (Nekoei and Weber, 2017; Lindner and Reizer, 2020).
- λ_U rises sharply (massive job destruction in services) (Autor et al., 2023).

8.3 Training cutoff: COVID

Hypothesis. The training cutoff rises (more workers train): the unskilled sector is disproportionately damaged, expanded UI lowers the cost of the training spell, making the skilled sector relatively more attractive (Blom et al., 2021; Ulvestad and Skjelbred, 2023). This provides a structural explanation for the observed enrolment spike.

8.4 Transition dynamics: COVID

Same objects as the financial-crisis application, starting from the pre-COVID steady state. Expect a sharp initial spike in unskilled unemployment with partial reversal as temporary layoffs are recalled.

9 Comparing the two crises

With both crises estimated, compare them systematically.

9.1 Parameter shifts

Side-by-side table of regime-specific parameter changes. Which parameters move in the same direction? Which move differently? Is the financial crisis primarily a “vacancy-cost shock” while COVID is a “separation-rate + UI shock”?

9.2 Training-cutoff comparison

$\bar{x}^{ss}(z_{FC})$ vs. $\bar{x}^{ss}(z_{COVID})$: opposite-sign predictions? Decompose contributions of each parameter shift to the change in the cutoff (Acabbi et al., 2024).

9.3 Transition-dynamics comparison

Overlay the two transition paths. Compare:

- Speed of adjustment.
- Overshooting / undershooting patterns.
- Which crisis generates longer-lasting hysteresis in the skilled share? (Acabbi et al., 2024; Audoly et al., 2022; Cohen et al., 2025).

10 Application 1: higher b_U and b_S —unemployment spell durations and match quality

This is a self-contained counterfactual exercise, separate from the regime-change analysis. We take one of the estimated baselines (pre-FC or pre-COVID) and ask how the stationary equilibrium responds to increases in the flow value of unemployment in either the unskilled sector (b_U), the skilled sector (b_S), or both.

10.1 Motivation

The flow value of unemployment is a central object in search models: it governs reservation productivities, acceptance behaviour, and—through free entry—equilibrium tightness (Pissarides, 2000; Shimer, 2005). A more generous unemployment benefit raises the worker’s outside option, which has two competing effects on welfare-relevant outcomes (Nekoei and Weber, 2017; Schmieder et al., 2016):

- **Duration effect.** A higher b_j raises reservation match quality p_j^* , so fewer contacts are accepted. Expected unemployment spell durations *rise*. Equilibrium tightness may also fall because free-entry vacancy creation responds to worker bargaining power (Hagedorn and Manovskii, 2008).
- **Match-quality effect.** Conditional on acceptance, accepted matches are of higher productivity (and higher wages). Average realised match quality *rises*, and subsequent endogenous separation rates may fall (Nekoei and Weber, 2017).

The structural model disciplines both margins simultaneously and delivers the general-equilibrium net effect. Because the unskilled and skilled markets differ in their shock processes, match-quality distributions, and on-the-job search, the two levers (b_U vs. b_S) may produce quantitatively and qualitatively different outcomes.

10.2 Exercise design

1. Fix all parameters at one of the estimated baselines.
2. Consider three policy arms:
 - **Arm U:** raise b_U by 10%, 25%, 50% (holding b_S fixed).
 - **Arm S:** raise b_S by 10%, 25%, 50% (holding b_U fixed).

- **Arm Joint:** raise both b_U and b_S proportionally.
- 3. For each arm and intensity, solve the new stationary equilibrium.
- 4. Compare outcomes across arms and intensities.

10.3 Outcomes to report

For each arm and intensity, report the new stationary-equilibrium values of:

- **Expected unemployment spell durations** $\mathbb{E}[D_U]$, $\mathbb{E}[D_S]$, and overall, computed from the equilibrium job-finding hazards $f_U(1 - F_U(p_U^*))$ and $f_S(1 - \Gamma_z(p_S^*))$.
- **Reservation match qualities** $p_U^*(x)$, $p_S^*(x)$: how much do acceptance thresholds rise?
- **Mean accepted match quality** in each sector (first moment of the equilibrium match-quality distribution among the employed).
- **Higher moments** of the match-quality distribution (variance, skewness): does the dispersion of realised matches change? (Lise et al., 2016).
- **Endogenous separation rates** δ_U , δ_S^{end} : higher reservation qualities mean workers start higher up the productivity distribution, so endogenous EU rates should fall.
- **Unemployment rates** ur_U , ur_S , ur_{total} : the net effect of longer durations and lower separations.
- **Market tightness** θ_U , θ_S : vacancy creation responds through free entry as worker surplus rises (Hagedorn and Manovskii, 2008).
- **Mean wages by sector**: higher reservation qualities feed into wages through bargaining.
- **Training cutoff $\bar{x}$ and training stock T** : changes in b_U vs. b_S move the option values of unskilled unemployment and skilled unemployment asymmetrically, so the training margin shifts differentially.
- **Welfare**: flow value of workers at various types x .

10.4 Discussion

- **Duration–match-quality trade-off.** Quantify the ratio of additional days unemployed per unit increase in average match quality. Is the trade-off more favourable in the unskilled or the skilled market? (Nekoei and Weber, 2017).
- **Asymmetry between b_U and b_S .** Raising b_S interacts with on-the-job search and the job ladder, so its effect on match quality operates both through the reservation threshold and through re-composition along the ladder (Cahuc et al., 2006; Bagger et al., 2014). Raising b_U additionally interacts with the training margin—a higher b_U makes remaining in the unskilled market more attractive, potentially *reducing* training take-up (Charlot and Decreuse, 2005).

- **General-equilibrium feedbacks.** Free-entry responses mean part of the partial-equilibrium gain in match quality is offset by lower tightness (Acemoglu, 2001; Hagedorn and Manovskii, 2008).
- **Dependence on the baseline.** Does the magnitude of these effects depend on whether we start from the pre-FC or pre-COVID baseline?

11 Application 2: education-subsidy policies

This is a self-contained counterfactual exercise, separate from the regime-change analysis. We take one of the estimated baselines (pre-FC or pre-COVID) and ask: how does the stationary equilibrium respond to two alternative education policies?

11.1 Motivation

The government wants to encourage training. Two policy levers are available, both targeting the training margin but through different channels (Card et al., 2018):

Policy A: higher training flow payoff b_T . The government provides monthly stipends to workers currently enrolled in training (e.g. living-cost grants, tuition waivers that effectively raise disposable income during the training spell). In the model, this is a direct increase in the flow value $b_T(z)$ while the worker is in the training state. It makes training more attractive by reducing the flow cost of the training spell (the foregone consumption during the period of non-employment) (Nekoei and Weber, 2017; Schmieder et al., 2016).

Policy B: lower training cost $c(x)$. The government invests in higher-education quality—better equipment, better instructors, improved facilities—at no additional cost to students. For a worker at ability x , the experience of training is now marginally better: the subjective disutility of studying is lower at every ability level. In the model, this is captured as a proportional reduction in the one-time cost function $c(x) = a_c e^{-x}$, i.e. replacing a_c with $(1 - \delta) a_c$ for some subsidy rate $\delta \in (0, 1)$ (Abbott et al., 2019; Heckman et al., 1998).

11.2 Key distinction between the two policies

- **Policy A** operates on the flow value during training. It makes the training state less painful on a per-period basis. It is temporary in the sense that the transfer stops once training completes.
- **Policy B** operates on the one-time entry cost. It lowers the up-front hurdle to initiating training. It affects the threshold at which a worker is indifferent between searching and training.

Both policies shift the training cutoff $\bar{x}(z)$ —more workers will choose to train—but they do so through different channels. Policy A raises the value of the training state $T(x, z)$ by increasing b_T ; Policy B lowers the cost $c(x)$ at every x , making the training option $U_U^{\text{train}}(x, z) = -c(x) + T(x, z)$ more attractive.

11.3 Exercise design

1. Fix all parameters at one of the estimated baselines.
2. For a range of subsidy intensities (e.g. 10%, 25%, 50% increase in b_T for Policy A; 10%, 25%, 50% reduction in a_c for Policy B), solve the new stationary equilibrium.
3. Compare outcomes across policies and across intensities.

11.4 Outcomes to report

For each policy and intensity level, report the new stationary-equilibrium values of:

- **Training cutoff $\bar{x}$:** how far does the cutoff shift? Which policy moves it more per dollar of government expenditure?
- **Training stock T :** how many additional workers are in training?
- **Skilled share $N_S/(N_U + N_S)$:** long-run change in skill composition.
- **Unemployment rates ur_U , ur_S , and ur_{total} :** does encouraging training reduce overall unemployment, or does it create congestion in the skilled market? (Acemoglu and Shimer, 1999).
- **Market tightness θ_U , θ_S :** how does vacancy creation respond? Does the unskilled market tighten (fewer searchers) while the skilled market loosens (more entrants)?
- **Mean wages by sector:** does the inflow of newly trained workers compress the skilled wage premium? (Bárány, 2016; Lee and Seshadri, 2019).
- **Separation and job-finding rates:** do equilibrium transition rates change?
- **Welfare:** flow value of workers at various types x .

11.5 Discussion

- **Which policy is more effective per unit of fiscal cost?** Policy A has a recurring cost (stipends paid every period for every trainee), while Policy B has a one-time capital cost (investing in institutions) that permanently lowers $c(x)$ (Abbott et al., 2019).
- **Are there diminishing returns?** At high subsidy rates, does one policy saturate faster than the other?
- **General-equilibrium feedbacks:** does the inflow of trained workers into the skilled market depress skilled wages or tighten the skilled market enough to partially offset the policy? (Heckman et al., 1998; Wasmer, 2006).
- **Does the ranking of the two policies depend on the current state of the economy** (i.e. which baseline we use)? This is a lighter interaction with the regime analysis—not a full regime-change exercise, just a comparison of which baseline produces which ranking.

12 Conclusion

Key takeaways.

- Training decisions respond endogenously to labour-market conditions, and the direction of the response depends on the nature of the shock (Dellas and Sakellaris, 2003; Blom et al., 2021).
- A demand shock (financial crisis) and a supply/reallocation shock (COVID) operate through different channels, leading to opposite predictions for training take-up (Bičáková et al., 2021, 2023).
- Cyclical shocks affect long-run skill composition through hysteresis in the training margin (Acabbi et al., 2024; Audoly et al., 2022).
- UI generosity in the unskilled and skilled markets generates a quantifiable trade-off between longer unemployment spells and higher realised match quality, with asymmetric effects on the training margin (Nekoei and Weber, 2017; Schmieder et al., 2016).
- Education subsidies can take the form of flow transfers (stipends) or one-time cost reductions (institutional investment), and the structural model can rank them by effectiveness (Card et al., 2018; Abbott et al., 2019).
- Structural models that nest regime changes and policy interventions provide a framework for crisis-contingent and investment-oriented labour-market policy.

References

- Abbott, Brant, Giovanni Gallipoli, Costas Meghir, and Giovanni L. Violante, “Education Policy and Intergenerational Transfers in Equilibrium,” *Journal of Political Economy*, 2019, 127 (6), 2569–2624.
- Acabbi, Edoardo Maria, Andrea Alati, and Luca Mazzone, “Human Capital Ladders, Cyclical Sorting, and Hysteresis,” 2024. Working Paper.
- Acemoglu, Daron, “Good Jobs versus Bad Jobs,” *Journal of Labor Economics*, 2001, 19 (1), 1–21.
- and Robert Shimer, “Holdups and Efficiency with Search Frictions,” *International Economic Review*, 1999, 40 (4), 827–849.
- Agopsowicz, Andrew, “Education and Search,” 2016. Working Paper.
- Audoly, Richard, Federica De Pace, and Giulio Fella, “Job Ladder, Human Capital, and the Cost of Job Loss,” Staff Reports 1043, Federal Reserve Bank of New York Dec 2022.
- Autor, David, Arindrajit Dube, and Annie McGrew, “The Unexpected Compression: Competition at Work in the Low Wage Labor Market,” 2023. NBER Working Paper No.

31010.

Bagger, Jesper, Fran

ccois Fontaine, Fabien Postel-Vinay, and Jean-Marc Robin, “Tenure, Experience, Human Capital, and Wages,” *American Economic Review*, 2014, *104* (6), 1551–1596.

Bárány, Zsófia L., “The Minimum Wage and Inequality: The Effects of Education and Technology,” *Journal of Labor Economics*, 2016, *34* (1), 237–274.

Barr, Andrew and Sarah Turner, “Out of Work and Into School: Labor Market Policies and College Enrollment during the Great Recession,” *Journal of Public Economics*, 2015, *124*, 63–73.

Bičáková, Alena, Guido Matias Cortes, and Jacopo Mazza, “Caught in the Cycle: Economic Conditions at Enrolment and Labour Market Outcomes of College Graduates,” *Economic Journal*, 2021, *131* (638), 2383–2412.

—, —, —, and —, “Make Your Own Luck: The Wage Gains from Starting College in a Bad Economy,” *Labour Economics*, 2023, *84*, 102411.

Blom, Erica, Brian C. Cadena, and Benjamin J. Keys, “Investment over the Business Cycle: Insights from College Enrollment,” *Journal of Political Economy*, 2021, *129* (12), 3304–3347.

Cahuc, Pierre, Fabien Postel-Vinay, and Jean-Marc Robin, “Wage Bargaining with On-the-Job Search: Theory and Evidence,” *Econometrica*, 2006, *74* (2), 323–364.

Card, David, Jochen Kluve, and Andrea Weber, “What Works? A Meta Analysis of Recent Active Labor Market Program Evaluations,” *Journal of the European Economic Association*, 2018, *16* (3), 894–931.

Charlot, Olivier and Bruno Decreuse, “Self-Selection in Education with Matching Frictions,” *Labour Economics*, 2005, *12* (2), 251–267.

— and —, “Over-Education for the Rich, Under-Education for the Poor: A Search-Theoretic Microfoundation,” *Labour Economics*, 2010, *17* (6), 886–896.

— and **Franck Malherbet**, “Education and Self-Selection in a Search Economy,” *Research in Economics*, 2013, *67* (2), 143–160.

—, **Bruno Decreuse**, and **Pierre Granier**, “Adaptability, Productivity, and Educational Incentives in a Matching Model,” *European Economic Review*, 2005, *49* (4), 1007–1032.

Clark, Damon, “Do Recessions Keep Students in School? The Impact of Youth Unemployment on Enrolment in Post-Compulsory Education in England,” *Economica*, 2011, *78*, 523–545.

Cohen, Jonathan, Andrew C. Johnston, and Attila Lindner, “Skill Depreciation during Unemployment: Evidence from Panel Data,” *American Economic Journal: Applied Economics*, 2025, *17* (3), 208–235.

- Decreuse, Bruno and Pierre Granier**, “Education, Matching Frictions and the Labor Market,” 2013. Working Paper.
- Dellas, Harris and Plutarchos Sakellaris**, “On the Cyclicity of Schooling: Theory and Evidence,” *Oxford Economic Papers*, 2003, 55 (1), 148–172.
- Diamond, Peter**, “Wage Determination and Efficiency in Search Equilibrium,” *Review of Economic Studies*, 1982, 49, 217–227.
- Flinn, Christopher, Ahu Gemici, and Steven Laufer**, “Search, Matching and Training,” 2016. Working Paper.
- Hagedorn, Marcus and Iourii Manovskii**, “The Cyclical Behavior of Equilibrium Unemployment and Vacancies Revisited,” *American Economic Review*, 2008, 98 (4), 1692–1706.
- Heckman, James, Lance Lochner, and Christopher Taber**, “Explaining Rising Wage Inequality: Explorations with a Dynamic General Equilibrium Model of Labor Earnings with Heterogeneous Agents,” *Review of Economic Dynamics*, 1998, 1 (1), 1–58.
- Lee, Sang Yoon (Tim) and Ananth Seshadri**, “On the Intergenerational Transmission of Economic Status,” *Journal of Political Economy*, 2019, 127 (2), 855–921.
- Leibing, Andreas**, “Skill Substitution, Expectations, and the Business Cycle,” 2026. Working Paper, TU Dresden.
- Lindner, Attila and Bal**
- ’azs Reizer**, “Frontloading the Unemployment Benefit: An Empirical Assessment,” *American Economic Journal: Applied Economics*, 2020, 12 (3), 1–31.
- Lise, Jeremy and Jean-Marc Robin**, “The Macro-dynamics of Sorting between Workers and Firms,” *American Economic Review*, 2017, 107 (4), 1104–1135.
- , **Costas Meghir, and Jean-Marc Robin**, “Matching, sorting and wages,” *Review of Economic Dynamics*, 2016, 19, 63–87. Special Issue in Honor of Dale Mortensen.
- Mortensen, Dale and Christopher Pissarides**, “Job Creation and Job Destruction in the Theory of Unemployment,” *Review of Economic Studies*, 1994, 61, 397–415.
- Moscarini, Giuseppe and Fabien Postel-Vinay**, “Wage Posting and Business Cycles,” *American Economic Review: Papers and Proceedings*, 2016, 106 (5), 333–339.
- and – , “On the Job Search and Business Cycles,” *Revue économique*, January 2024, 75 (1), 73–112.
- Nekoei, Arash and Andrea Weber**, “Does Extending Unemployment Benefits Improve Job Quality?,” *American Economic Review*, 2017, 107 (2), 527–561.
- Pissarides, Christopher A.**, *Equilibrium Unemployment Theory*, 2 ed., Cambridge, MA: MIT Press, 2000.

- Postel-Vinay, Fabien and Jean-Marc Robin**, “Equilibrium Wage Dispersion with Worker and Employer Heterogeneity,” *Econometrica*, 2002, 70 (6), 2295–2350.
- Robin, Jean-Marc**, “On the Dynamics of Unemployment and Wage Distributions,” *Econometrica*, 2011, 79 (5), 1327–1355.
- Schmieder, Johannes F., Till von Wachter, and Stefan Bender**, “The Effect of Unemployment Benefits and Nonemployment Durations on Wages,” *American Economic Review*, March 2016, 106 (3), 739–77.
- Shimer, Robert**, “The Cyclical Behavior of Equilibrium Unemployment and Vacancies,” *American Economic Review*, 2005, 95 (1), 25–49.
- Ulvestad, Marte E. S. and Siv-Elisabeth Skjelbred**, “Are Graduate School Applications Countercyclical? Evidence from Norway,” *Economics of Education Review*, 2023, 94, 102404.
- Wasmer, Etienne**, “General versus Specific Skills in Labor Markets with Search Frictions and Firing Costs,” *American Economic Review*, 2006, 96 (3), 811–831.